

A mass-accurate machine-learning surrogate for molecular hydrogen density at unseen UV field strengths

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ABSTRACT

Can a machine-learning surrogate predict the molecular hydrogen density field of an interstellar-medium simulation at a UV field strength it has never seen — accurately enough to trust the prediction as a *physical* field? We test this on seven 128^3 UV-only chemistry simulations spanning $G_0 = 0.1$ – 6.4 Habing, holding out one entire simulation per fold and scoring predictions by H_2 mass budget, bias, and phase structure as well as log-space R^2 . Three ingredients suffice: multi-scale box-filter neighbourhood features (a cheap shielding proxy) halve the residual scatter of tabular models; density-weighted training and a mass-weighted bias calibration, fitted on training cubes alone, remove a $+0.3$ dex systematic offset that would otherwise inflate the H_2 mass by 70–90 per cent. The resulting pipeline attains $\text{RMSE} = 0.25$ dex, $R^2 = 0.990$, and total H_2 mass within 9 per cent on every held-out fold. A 3D U-Net is more accurate when interpolating but over-predicts the mass of the hardest extrapolation fold by more than two orders of magnitude — a failure invisible to R^2 , which separates the two models by only 0.02. Physically scored, out-of-distribution reliability is set by input representation and calibration, not architecture.

Key words: ISM: molecules – ISM: clouds – astrochemistry – methods: numerical – methods: statistical

1 INTRODUCTION

The H_2 abundance sets the initial conditions of star formation, but computing it self-consistently in a three-dimensional simulation means integrating stiff non-equilibrium chemistry — grain-surface formation (Cazaux & Tielens 2004), Lyman–Werner photodissociation (Habing 1968; Draine & Bertoldi 1996), and self-shielding (Wolcott-Green et al. 2011) — in every cell, a cost that limits resolution and parameter coverage (Grassi et al. 2014). Machine-learned emulators of ISM chemistry (de Mijolla et al. 2019; Palud et al. 2023; Holdship et al. 2021; Grassi et al. 2022; Branca & Pallottini 2024; Janssen et al. 2026) replace the solver with a fast local mapping, but they are zero-dimensional — each cell is treated independently — while the equilibrium n_{H_2} of a cell depends on the shielding column of gas around it, and they are conventionally scored by log-space error statistics, while their downstream uses (mass budgets, phase statistics, synthetic observations) consume the prediction as a physical field.

This paper asks one question: *can a surrogate trained on simulations at six UV field strengths predict the full n_{H_2} field of a seventh simulation, at a field strength absent from training, accurately enough to be used as a physical field?* The question fixes the design. Generalisation is measured by leave-one- G_0 -out cross-validation over a controlled seven-simulation UV sweep — every fold withholds an entire 128^3 cube. Success is measured by a metric hierarchy in which the primary statistics are physical: the RMSE decomposed exactly into bias and scatter, and the total H_2 mass ratio; log-space R^2 is retained but demoted to a secondary statistic. Non-locality is treated as

a representation problem: we compare engineered multi-scale neighbourhood features attached to tabular models (gradient-boosted trees and multilayer perceptrons) against a volumetric 3D U-Net (Ron-neberger et al. 2015; Çiçek et al. 2016) that can learn shielding-like aggregations through its receptive field.

The answer is yes, with qualifications that only the physical scoring reveals. The best least-squares model carries a $+0.32$ dex systematic offset on held-out cubes — invisible at the precision R^2 is quoted, but a 70–90 per cent error in total H_2 mass. Repairing it requires treating the bias as a modelling target of its own: a two-parameter, mass-weighted calibration fitted on training cubes alone closes the mass budget in every fold. The U-Net, conversely, is the best interpolator in the comparison and fails at both extrapolation boundaries in a phase-localised, non-monotonic way that no global calibration can repair. Section 2 describes the data, protocol, and metrics; Section 3 the models; Section 4 the results; Section 5 what they imply.

2 DATA, PROTOCOL, AND METRICS

2.1 Simulation suite

The data are seven 128^3 uniform-grid simulations (14.7 million cells in total) evolving identical turbulent ISM initial conditions under a spatially uniform, time-steady UV field of strength $G_0 \in \{0.1, 0.2, 0.4, 0.8, 1.6, 3.2, 6.4\}$ Habing, illuminating each cube along $+z$; the snapshots are quasi-equilibrium chemical states. The suite is a controlled single-parameter sweep: the runs share one turbulent realisation, so all generalisation claims concern the UV-field axis alone (Section 5), though the evolved snapshots differ hydrodynamically as well as chemically (the median temperature rises

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by an order of magnitude across the sweep). The target n_{H_2} spans ≈ 16 orders of magnitude and its per-cube mean falls from -3.1 to -7.2 in $\log_{10}$ as photodissociation strengthens, so all modelling is done in $\log_{10}(n_{\text{H}_2})$ and each held-out cube has a target distribution shifted relative to everything in training.

Each cell provides 15 input features: $\log n_{\text{H}}$, $\log T$, $\log n_{\text{H}^+}$, a dust-extinction proxy, $\log G_0$, three velocity and six face-centred magnetic-field components, and the solver’s H_2 self-shielding factor $\log f_{\text{H}_2}$. The last is special: it is a column integral over the very field being predicted, computed by the solver the surrogate aims to replace, and would not be available in deployment. We therefore quantify its contribution by ablation (Section 4.5) and regard the configuration without it as the deployable one.

2.2 Protocol and metrics

All headline results use leave-one- G_0 -out seven-fold cross-validation; every model, tabular or volumetric, is scored on the same native 128^3 cells of the held-out cube. Random cell-level splits would put every G_0 and the shared realisation in both train and test — a pointwise tree already exceeds $R^2 = 0.99$ there — and would say nothing about the intended use, prediction at an unsimulated field strength. The two boundary folds ($G_0 = 0.1, 6.4$) require genuine extrapolation; the five interior folds require interpolation. Single-cube transfer experiments confirm the protocol is non-trivial: a model trained on one cube reaches mean $R^2 = 0.966$ one geometric step away in G_0 but only 0.888 at two steps and essentially zero at the largest separations, so no training cube is redundant.

Primary metrics. The RMSE in $\log_{10}(n_{\text{H}_2})$, decomposed exactly as $\text{RMSE}^2 = b^2 + s^2$ with b the mean residual (bias) and s the residual scatter; and the total H_2 mass ratio $\mathcal{M} = \sum_i 10^{\hat{y}_i} / \sum_i 10^{y_i}$, the ratio of predicted to true H_2 mass on the uniform grid. A bias of b dex, invisible to variance-normalised statistics, multiplies the mass budget by up to 10^b .

Secondary metrics. Log-space R^2 (for comparability with the emulator literature); phase-conditional R^2 and bias for the molecular ($y > -4$) and diffuse ($y \leq -4$) populations, the threshold lying between the two modes of the target distribution and the conclusions insensitive to its placement within $[-5, -3]$; the fraction $f_{0.1}$ of cells within 0.1 dex; the mass-weighted MAE; the Wasserstein distance W_1 between predicted and true marginals; and a skill score $1 - \text{MSE}/\text{MSE}_{\text{XGB}}$ against the pointwise gradient-boosted baseline. Folds, not cells, are the inferential unit; cross-fold means carry the fold standard deviation, which bootstrap resampling shows dominates the within-fold sampling uncertainty by an order of magnitude.

Before exponentiation, predictions are clipped to within 1 dex of the held-out fold’s true range so a handful of outlier cells cannot dominate $\mathcal{M}$; for well-calibrated models the clip is never active, and where it is we report $\mathcal{M}$ as a lower bound and flag it. Every experiment writes a timestamped JSON log (configuration, seeds, per-fold metrics, data checksum); tree results are bit-identical across runs.

3 MODELS

3.1 Spatial features, weighting, and calibration

Multi-scale neighbourhood features. To give per-cell models access to spatial context we precompute box-filter averages of all 15 input fields at kernel widths $k \in \{3, 5, 7\}$ voxels (reflected at box faces), concatenated with the local features into a 60-dimensional vector.

The separable filter costs $O(N)$ per scale; all 45 filtered fields for all seven cubes take ≈ 30 s. Averages of the column-type fields (density, extinction) act as crude shielding estimators.

Density-weighted training. The cell population is dominated by diffuse gas while the H_2 mass resides in the densest few per cent of cells, so an unweighted log-space loss treats the mass-carrying tail as an afterthought. Weighted variants use smooth exponential sample weights rising from $1\times$ at the 99th percentile of the training targets to $100\times$ at the 99.99th, normalised to unit mean.

Stacking and bias calibration. A Ridge ($\alpha = 1$) meta-learner stacks the spatial-feature XGBoost and MLP, whose inductive biases are complementary (bounded piecewise-constant extrapolation versus smooth interpolation). Stacked predictions minimise squared error but are not unbiased on an unseen G_0 (Section 4.2), so for each fold we fit the per-training-cube stack residual against $\log G_0$ by two-parameter least squares — training-cube quantities only — and subtract the fitted value at the held-out G_0 . The residual functional is a physical choice we evaluate explicitly: *mean* calibration centres the unweighted per-cube mean log residual; *mass* calibration centres the mass-weighted mean residual $b_M = \sum_i 10^{y_i} (\hat{y}_i - y_i) / \sum_i 10^{y_i}$, the constant offset that approximately closes the total-mass budget (on the deployed volumes it agrees with exact per-cube mass closure to within 0.018 dex in every fold).

3.2 Model families

XGBoost (Chen & Guestrin 2016): 400 rounds, depth 6, GPU histogram method; depth variants change mean R^2 by < 0.003 . *Wide MLP*: hidden widths [512, 512, 256, 128], batch normalisation, Adam, 100 epochs per fold; predictions clamped to the training-target range ± 2 dex; architecture variants agree to within seed-to-seed variation. *3D U-Net*: three encoder stages with residual blocks and instance normalisation, trained on whole 128^3 volumes (batch size 1, 150 epochs, unweighted loss, unbounded output) with the eight z -preserving symmetry operations as augmentation — the UV illumination axis breaks the remaining 40 octahedral operations, and using them degrades accuracy. The held-out cube is never augmented. Its checkpoint is selected per fold on an inner validation cube drawn from the six training cubes (the cube closest to the median $\log G_0$, keeping every fold’s adjacent- G_0 neighbour in training), so no held-out information enters model selection; the selection is nearly a no-op against simply taking the final epoch, though the choice of validation cube visibly moves U-Net numbers (mean R^2 0.914–0.963 across the two rules we evaluated). In the comparison table the stack meta-learner reuses out-of-fold predictions from the outer cross-validation (the standard shortcut); the deployed pipeline of Section 4.3 is fully nested and scores slightly *higher*, so the table rows are conservative.

4 RESULTS

4.1 What each ingredient buys

Table 1 and Figure 1 carry the main result; the bias–scatter decomposition assigns each ingredient a distinct role.

Spatial features buy precision. The 45 neighbourhood features reduce XGBoost’s RMSE from 0.49 to 0.37 dex and the MLP’s from 0.52 to 0.32 dex, almost entirely through the scatter term (0.36 to 0.25 and 0.46 to 0.29 dex respectively) while the bias barely moves: non-local context sharpens the prediction, it does not recalibrate it.

Density weighting buys the mass budget, free. The weighted tree is indistinguishable from the unweighted one in R^2 yet its fold mass

Table 1. Leave-one- G_0 -out performance (cross-fold means; mass ratio M as the min–max fold range). “+S” appends the 45 spatial features; “+W” density-weighted training; stack rows are the Ridge-stacked tree–MLP ensemble raw and with the two calibrations of Section 3.1. $\text{RMSE}^2 = b^2 + s^2$ exactly. R_{mol}^2 is the molecular-phase ($\log_{10}(n_{H_2}) > -4$) R^2 ; skill is measured against the pointwise XGBoost row. † Lower bound: predictions clipped at truth +1 dex before the mass integral (Section 2.2). The final row is the deployed fully nested pipeline (Section 4.3), for which the clip is never active.

Model	$R^2 (\pm \sigma)$	RMSE (dex)	bias b (dex)	scatter s (dex)	M (range)	R_{mol}^2	skill
XGBoost	0.950 ± 0.029	0.486	+0.24	0.36	0.51–0.90	0.978	0.00
Wide MLP	0.958 ± 0.028	0.518	+0.21	0.46	1.00–2.59	0.981	−0.18
XGBoost +S	0.963 ± 0.033	0.374	+0.20	0.25	0.47–0.93	0.988	+0.32
Wide MLP +S	0.982 ± 0.017	0.322	+0.11	0.29	0.94–1.16	0.989	+0.46
XGBoost +S+W	0.963 ± 0.033	0.374	+0.20	0.25	0.89–1.03	0.989	+0.32
Wide MLP +S+W	0.981 ± 0.020	0.329	+0.10	0.30	0.92–1.03	0.989	+0.41
Stack (+S+W), raw	0.959 ± 0.026	0.433	+0.32	0.26	1.65–1.84	0.955	+0.02
Stack (+S+W), mean-cal.	0.984 ± 0.011	0.301	−0.01	0.26	0.71–0.89	0.981	+0.57
Stack (+S+W), mass-cal.	0.985 ± 0.015	0.294	+0.09	0.26	0.93–1.07	0.991	+0.55
3D U-Net	0.963 ± 0.047	0.348	−0.09	0.29	0.71–291 †	0.74	+0.36
Deployed stack, nested, mass-cal.	0.990 ± 0.006	0.250	+0.02	0.24	0.99–1.09	0.990	+0.74

ratios tighten from 0.47–0.93 to 0.89–1.03; the weighted MLP behaves the same way. Upweighting the mass-carrying tail fixes *where* the model is accurate at no aggregate cost.

Stacking buys squared error and exposes a bias. The Ridge stack has the lowest scatter in the comparison but a +0.32 dex mean offset on held-out cubes — the meta-learner’s intercept zeroes the residual on the pooled training folds, not on an unseen G_0 — inflating the H_2 mass by 65–84 per cent while still scoring $R^2 > 0.91$ in every fold. This is the central methodological observation: the model that minimises squared error is *not* the model whose field can be trusted physically, and R^2 cannot tell them apart. (R^2 also inverts rankings outright: the pointwise MLP beats pointwise XGBoost on mean R^2 while its pooled squared error is 18 per cent worse, because R^2 ’s fold-variance denominator rewards winning high-variance folds.)

4.2 Calibration closes the budget

The stack’s offset varies smoothly with the held-out G_0 , exactly the structure the two-parameter calibration removes; fitted corrections are +0.2 to +0.4 dex. The residual functional matters because the stack’s error is density-dependent (dense cells under-predicted relative to diffuse ones): *mean* calibration centres the cell-mean bias essentially perfectly ($b = -0.01$ dex, the best skill in the table) but overshoots the budget, dragging fold mass ratios to 0.71–0.89, while *mass* calibration closes the budget (0.93–1.07) at the cost of a +0.09 dex offset carried by the diffuse population. The 0.1–0.15 dex gap between the two fitted offsets is itself a one-number diagnostic of density-dependent error. Under the primary metrics the mass-calibrated weighted stack is the best configuration — lowest RMSE of all variants evaluated, mass closed in every fold, molecular-phase $R^2 = 0.991$ — and we adopt it; applications weighting all cells equally may prefer the mean-calibrated variant, and Table 1 reports both. The calibration is stable: refitting with each training cube removed in turn moves the applied offset by at most 0.04 dex.

4.3 The deployed pipeline

The deployed pipeline — the density-weighted spatial stack with fully nested stacking and mass calibration — reaches per-fold R^2 of 0.977–0.998 (mean 0.990 ± 0.006), RMSE 0.25 dex, cell-mean bias +0.02 dex, total H_2 mass within 0.985–1.090 of truth in *every* fold, molecular-phase $R^2 = 0.979$ –0.994, mass-weighted MAE 0.06 dex, and mean $W_1 = 0.08$ dex between predicted and true marginals

(Figure 2). The metric-level clip is never active for these volumes. Residual errors concentrate in the diffuse UV-exposed gas at low G_0 , where small shielding-column differences move cells across the steep atomic-to-molecular transition; the hardest fold is the *interior* one at $G_0 = 0.8$ (bias +0.16 dex), a reminder that interpolation folds are not automatically easy. The full pipeline — features, training, calibration, prediction for all seven folds — runs in about an hour on one consumer GPU and predicts a new 128³ cube in seconds.

4.4 The volumetric alternative

The U-Net occupies a revealing position: it is simultaneously the best interpolator and the least reliable extrapolator. On the five interior folds it is the most accurate model evaluated ($R^2 = 0.992$, RMSE 0.22 dex, against 0.984 and 0.31 dex for the calibrated stack) — direct evidence that a learned volumetric representation extracts non-local information the box filters miss. At the boundaries the picture inverts: mean edge-fold R^2 falls to 0.889 (RMSE 0.66 dex), at $G_0 = 0.1$ it is far worse than the pointwise XGBoost baseline (skill −1.37), and at $G_0 = 6.4$ its molecular-phase bias is +1.13 dex with a fold mass ratio above 290 (a lower bound under the clip). Crucially, this failure is not calibration-repairable: the U-Net’s held-out bias is phase-localised and non-monotonic in G_0 (fold mass ratios 1.7, 2.1, 1.1, 0.71, 1.8, 19.0, 291), whereas the stack’s bias is a global, G_0 -smooth offset — the difference between a correctable systematic and a genuine extrapolation breakdown. The U-Net is also $\sim 5\times$ costlier to train than the entire tabular comparison and, with 48 augmented volumes against millions of parameters, configuration-sensitive (mean R^2 0.80–0.97 across capacity and regularisation variants in a dedicated study). We did not pursue CNN-specific remedies (weighted volumetric losses, bounded outputs, pretraining), so we make no claim about the ultimate capability of volumetric architectures; on a suite of this size, engineered spatial context plus calibration is the robust default, and the volumetric model is preferable only within a densely sampled parameter range.

4.5 Dependence on the solver’s shielding factor

One input, f_{H_2} , is produced by the solver the surrogate replaces, so its contribution must be quantified, not assumed. Excluded from both the local features and the spatial averages, the deployed configuration retains $R^2 = 0.964 \pm 0.014$ with its mass budget intact (fold ratios 0.93–1.16) — but its RMSE grows by 60 per cent (0.29 to 0.47 dex)

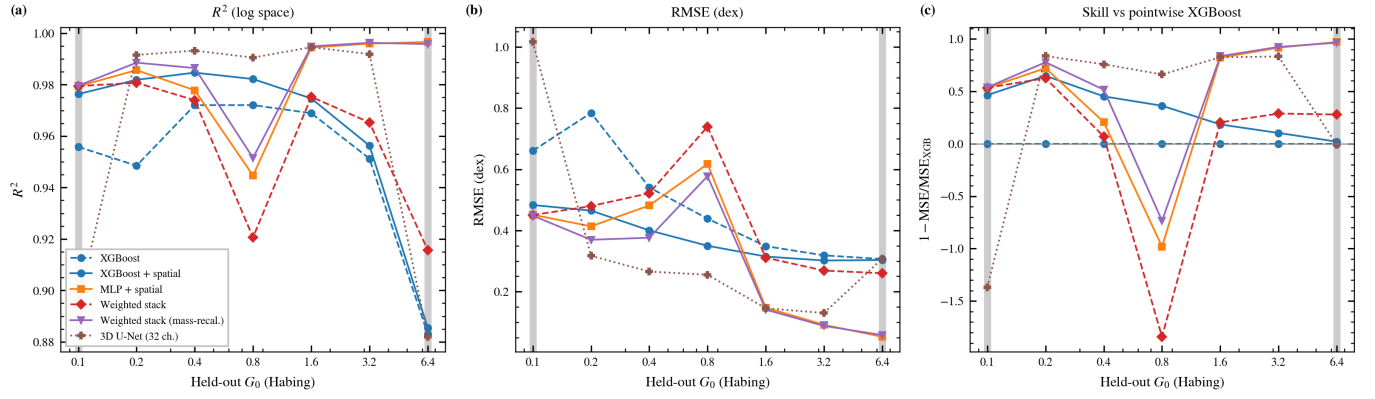


Figure 1. Per-fold behaviour across the G_0 sweep: log-space R^2 (a), RMSE (b), and skill against the pointwise XGBoost baseline (c). Grey bands mark the two extrapolation folds. The 3D U-Net (dotted) is the most accurate model on the interior folds and the least robust at both boundaries; the mass-calibrated weighted stack (purple) is the only model that remains strong across the entire range.

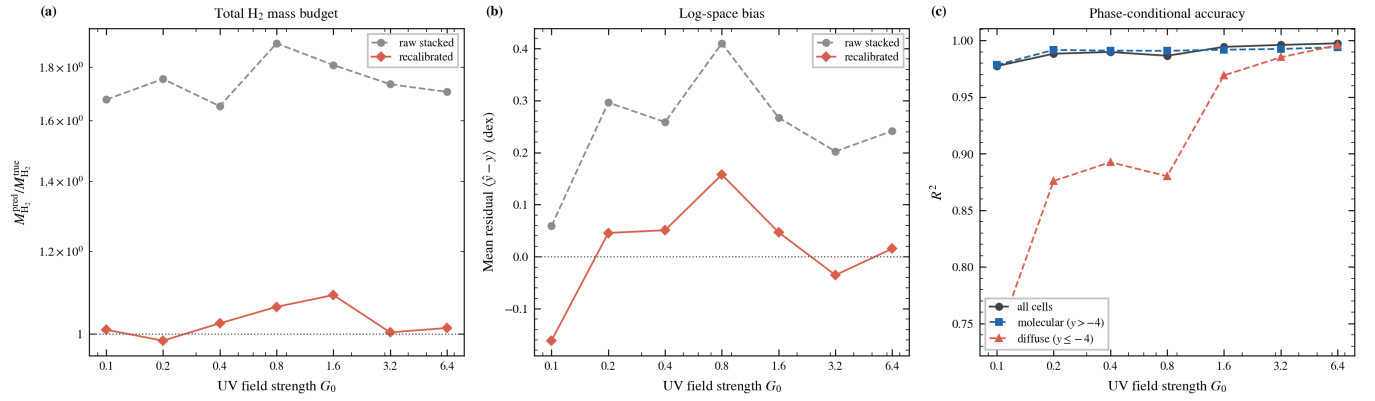


Figure 2. The deployed pipeline, per held-out fold. (a) Total H_2 mass ratio: the raw stack (grey) over-predicts by $\times 1.65$ – 1.90 at every G_0 ; the mass-calibrated prediction (red) is within 9 per cent of the true mass in every fold. (b) Cell-mean log residual. (c) Phase-conditional R^2 : molecular-phase accuracy is uniformly high; the diffuse phase is hardest at low G_0 , where UV-exposed gas spans the steep atomic-to-molecular transition.

and the molecular-phase R^2 falls from 0.991 to 0.840, reaching 0.70 in the most molecular cube ($G_0 = 0.1$). The spatial features absorb roughly half of the pointwise models’ ablation loss, consistent with neighbourhood averages of density and extinction acting as crude column estimators, and XGBoost gain importance corroborates the picture (log f_{H_2} carries 0.57 of the pointwise gain, 0.83 with its spatial averages; velocity and magnetic-field components are negligible — a rerun without them leaves XGBoost unchanged and the calibrated stack at $R^2 = 0.975$ with its mass budget closed, though the MLP branch destabilises in log space). The complementary bound comes from a U-Net trained on 11 chemistry-free inputs (temperature, G_0 , velocity, magnetic field): it still reaches $R^2 = 0.851 \pm 0.121$ — the H_2 morphology is largely inferable from dynamics, thermodynamics, and the UV strength — but with 0.72 dex scatter, near-zero molecular-phase R^2 , and fold mass ratios from 0.29 to a clipped $\times 510$. Morphology is cheap; chemistry is not. Deployable accuracy *inside* the molecular phase therefore depends on a shielding estimate of f_{H_2} quality, e.g. from extinction or an approximate radiative-transfer sweep.

5 DISCUSSION AND CONCLUSIONS

The question posed in Section 1 has a quantitative answer: a per-cell surrogate with engineered neighbourhood features, density-weighted training, and a mass-weighted bias calibration predicts the n_{H_2} field of an unseen- G_0 simulation with RMSE 0.25 dex, $R^2 = 0.990$, and total H_2 mass within 9 per cent in every fold — and each of those three ingredients addresses a distinct error component (scatter, mass placement, bias). Two broader lessons follow.

First, *scoring determines conclusions*. Ranked by log-space R^2 , the raw stack, the calibrated stacks, and the U-Net are nearly interchangeable (0.959–0.985); ranked by the physical metrics they span mass errors from 7 per cent to two orders of magnitude. The bias-scatter decomposition, the mass ratio, and phase-conditional errors should be standard reporting for field-level emulators — and the choice of calibration functional (cell-mean versus mass-weighted) is a physical decision that changes the recovered mass by tens of per cent, so it should be stated and, ideally, both variants reported.

Second, *representation and calibration beat architecture out-of-distribution*. The spatial features cut tabular RMSE by 23–38 per cent — several times more than any architecture variation we tested, consistent with the competitiveness of tree ensembles on tabular data (Grinsztajn et al. 2022) — while the U-Net’s genuine interpolation

advantage does not survive leaving the training range. For suites of this size the practical division of labour is: engineered features plus calibration for robust extrapolation; the volumetric model only within a densely sampled parameter range.

The claims are bounded by the suite’s design. The seven simulations share one turbulent realisation, so generalisation across realisations, metallicities, dust models, or resolutions is untested; the suite is quasi-equilibrium and uniformly illuminated; the box-filter features and the U-Net’s zero padding approximate the context of cells near box faces; and the density-weighting anchors are a designed choice whose sensitivity we have not mapped. The deployable (no- f_{H_2}) configuration is a usable surrogate for aggregate statistics and morphology, but per-cell molecular-phase accuracy requires genuine shielding information. Within those bounds the result stands: judged as a physical field, the H_2 density of an unseen UV environment is predictable — provided the surrogate is scored, weighted, and calibrated in physical terms.

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AI tools were used during the project to assist with code generation and debugging, exploratory analysis of results, and drafting and editing of the manuscript. All scientific decisions, interpretation of results, validation of code outputs, and final manuscript content are the responsibility of the author.

SOFTWARE

Python 3.14.5; PyTorch 2.12 (Paszke et al. 2019); XGBoost 3.2 (Chen & Guestrin 2016); NumPy 2.4 (Harris et al. 2020); SciPy 1.17 (Virtanen et al. 2020); scikit-learn 1.8 (Pedregosa et al. 2011); Matplotlib 3.10 (Hunter 2007). All experiments were run on a single NVIDIA GeForce RTX 3090.

DATA AVAILABILITY

All experiment logs, trained-model outputs, and prediction volumes are archived as timestamped JSON and .npz files and are available from the author on request. The simulation suite is not yet public; requests for data access may be directed to the corresponding author.

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